Working Memory in a Recurrent Spiking Neural Networks With Heterogeneous Synaptic Delays

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Abstract. Working memory — the ability to store and recall precise temporal patterns of neural activity — remains an open challenge for spiking neural networks (SNNs). We propose a recurrent SNN of N neurons in which each synapse is equipped with $D = 41$ delays, modelled as a weight tensor $\mathbf{W} \in \mathbb{R}^{N \times N \times D}$ and trained end-to-end with surrogate-gradient backpropagation through time. The network stores M arbitrary target spike patterns by representing each as a sequential chain of overlapping Spiking Motifs: contiguous windows of length D that uniquely predict spikes at the next time step. On a synthetic benchmark of $M = 16$ patterns ($N = 1024$ neurons, $T = 1000$ steps), the network after training achieves a perfect F1 score of 1.0 for the prediction of each spike of the given patterns. This result demonstrates that heterogeneous delays provide an efficient substrate for working memory in SNNs, enabling energy-efficient neuromorphic edge deployment.

Keywords: Spiking Neural Networks · Heterogeneous delays · Working memory.

1 Introduction

A century ago, Adrian and Zotterman [2] established that neural communication relies on all-or-none action potentials, *spikes*. Whether information is encoded in precise spike timing or in the coarser measure of instantaneous firing rate remains debated [4, 10, 23] — a question that becomes especially sharp for working memory, where the brain must maintain and manipulate information across delays of seconds. In traditional artificial neural networks, long-range temporal dependencies are addressed by gated recurrent unit (GRU) or Transformers [25] architectures, which decouple temporal binding range from the timescale of individual units. The analogous problem for spiking neural networks (SNNs) — binding activity patterns separated by intervals far exceeding a single neuron's time constant — lacks a general solution and remains a key challenge for biologically plausible sequence processing [28].

One principled approach relies on the *precise relative timing* of spikes across a population. Abeles [1] was a precursor in proposing that cortical neurons

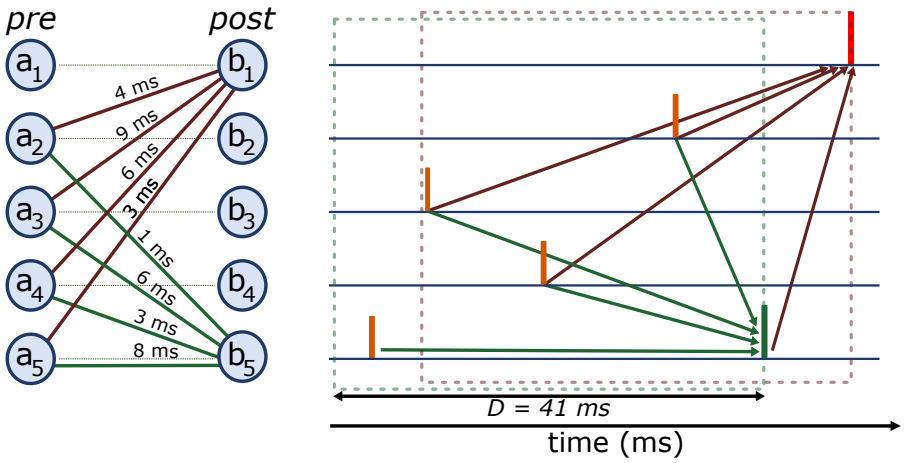

Fig. 1. Working memory as sequential spike prediction via heterogeneous delays. *Left:* A recurrent HD-SNN of five neurons numbered 1–5, connected by two subsets of recurrent connections that define two Spiking Motifs from the presynaptic side (neurons a) to the postsynaptic side (neurons b): dark-red connections from a_2, a_3, a_4, and a_5 target b_1 with delays 4, 9, 6, 3 ms; green connections from a_1, a_2, a_3, a_4 target b_5 with delays 1, 6, 3, 8 ms. *Right:* Raster plot illustrating how the network predicts successive spikes from a past context window. *Step 1 — first prediction (green dashed box):* the orange input spikes emitted at staggered times by a_2–a_5 within the first context window (dashed green box) with precise interspike intervals. The (green) delays of all four spikes converge *synchronously* at b_5 define a first Spiking Motif such that the spikes in the (green dashed) context window cause the (green) output spike at b_5 at a precise timing. *Step 2 — second prediction (dark-red dashed box):* the green spike of a_5 re-enters the network as a new input and, together with the spikes now visible in the shifted context window (dark-red dashed box), is routed through the dark-red delays in order to converge synchronously on b_1, generating a new output spike. Because output spikes re-enter the same population, detecting one Spiking Motif provides the context for predicting the next spike in the sequence, implementing working memory as a chain of temporal predictions grounded in the preceding context window.

act as coincidence detectors; Bienenstock [3] formalised this into *synfire chains*, feedforward pools in which synchronous volleys propagate reliably. Experimental support came from Bruno and Sakmann [5], who showed that thalamocortical inputs are individually weak but effective through precise synchrony; theoretical work further showed such synchrony can be learned in an unsupervised manner [19]. Ikegaya et al. [13] then directly observed recurring spatiotemporal spike sequences in neocortex at both sub- and supra-second timescales, and complementary hippocampal evidence shows that such sequences self-organise spontaneously as internal population templates [26]. The recent study by Xie et al. [27] demonstrates that in humans, information is using stereotypical spatiotemporal spike sequences within population bursts, complementing traditional rate- and latency-based neural codes.

Izhikevich [14] extended the synfire framework by introducing heterogeneous axonal delays in a recurrent population of spiking neurons trained with spike-timing-dependent plasticity (STDP). The resulting *polychronous neuronal groups* (PNGs) — time-locked, non-synchronous firing patterns at millisecond precision — vastly outnumber the neurons in the network, yielding a large short-term memory capacity exploited for visual feature binding [6] and sequence classification [18]. Szatmáry and Izhikevich [24] showed that short-term plasticity can cue a PNG's reactivation and sustain a working memory. However, storing spike patterns of arbitrary length is inherently difficult: the sequential structure of the task means each context window (that is, the range of neurons and time which may hit a neuron monosynaptically) is not an independent query but is itself the output of the network at the previous step, so small errors at early time steps propagate and amplify towards the end of the sequence — a compounding difficulty absent from static associative memory like Hopfield networks.

The limitation of the original model from Izhikevich [14] is that delays are fixed rather than learned. Recent work addresses this: Göltz et al. [8] demonstrated fast, energy-efficient inference on neuromorphic hardware using first-spike-time coding, while Hammouamri et al. [12] introduced gradient-based delay learning via dilated convolutions with learnable spacings (DCLS), achieving sparse solutions in which only $\sim 20\%$ of delay channels remain active. Applied to event-based vision, heterogeneous learned delays in a single spiking layer suffice for fast motion detection [11], scalable to always-on object recognition [9]. Instead of observing the emergence of PNGs, we define *Spiking Motifs* (SMs), precisely timed spike patterns that may be trained by tuning multiple learned delays per synapse in a back-propagation trainable, hardware-compatible framework, here referred to as Heterogeneous-Delay SNNs (HD-SNNs) [20]. Kronland-Martinet et al. [15] recently extended this to motifs of arbitrary length by chaining sub-motif detectors with bounded synaptic delays, achieving robust encoding of complete Spiking Heidelberg Digits patterns.

The present work generalises this chain-of-motifs framework to enable working memory in a recurrent SNN. Rather than relying on fixed delays and unsupervised STDP rules as in the original polychronisation model, we extend the spike-timing theory of [15] to a fully recurrent topology: once primed by a short "trigger"

window, the network autonomously recalls arbitrary-length spike sequences. The model is trained end-to-end with surrogate gradients [17], bridging the computational neuroscience framework of polychronisation with gradient-based machine learning and targeting the energy-efficiency and sparsity objectives central to neuromorphic edge deployment.

2 Methods

2.1 Representation of spiking activity

In neurobiological recordings or the output of event-based cameras, any generic raster plot consists of a stream of *spikes*, defined as a list of address–timestamp tuples

$$\mathcal{E} = \big\{(a_r,\, t_r)\big\}_{r\in[1,\, N_{\mathrm{ev}}]},$$

where $N_{\mathrm{ev}} \in \mathbb{N}$ is the total number of events and r indexes each event in temporal order [9]. Each event has a time of occurrence t_r and an address a_r in the neural population space $\mathcal{A}$. In a Single Unit Activity (SUA) recording for instance, $\mathcal{A}$ is the identified set of neurons.

Neurons are characterised by their membrane potential dynamics and by the integration of synaptic potentials along their dendritic tree. Crucially, neurons can receive inputs from multiple presynaptic neurons with *heterogeneous delays* δ_s, representing the travel time of the presynaptic spike to the soma of the postsynaptic neuron. Scanning all pre- and postsynaptic pairs, we define the set of $N_s \in \mathbb{N}$ synapses as

$$\mathcal{S} = \big\{(a_s,\, b_s,\, w_s,\, \delta_s)\big\}_{s\in[1,\, N_s]},$$

where $a_s \in \mathcal{A}$ and $b_s \in \mathcal{A}$ are respectively the pre- and postsynaptic addresses, w_s is the synaptic weight, and δ_s is the delay. Note that neurons in the human cortex have on average 30000 synapses [22], an order of magnitude which is seldom reached in artificial networks. In our model, two neurons may contact each other via multiple synapses, each with a different delay. The *receptive field* of neuron b is the sub-set of synapses that project onto it:

$$\mathcal{S}^b = \big\{(a_s,\, b_s,\, w_s,\, \delta_s) \mid b_s = b\big\}_{s\in[1,\, N_s]}.$$

Each spike from the event stream $\mathcal{E}$ gets multiplexed in time by these synapses to later converge on the post-synaptic neurons. Crucially, as they reach the post-synaptic neuron, a synchronous convergence of multiple spikes on the soma — made possible by the interplay of heterogeneous delays — increases the post-synaptic firing probability, enabling the detection of precise spiking motifs (see Figure 1).

To allow for the simulation of this network on standard hardware we transform this event-based representation into the Boolean matrix $A \in \{0,1\}^{N\times T}$, where $N = |\mathcal{A}|$ is the number of neurons and T is the number of discrete time steps of

1 ms each (see Figure 2, right panel). Delays are discretised as integers in $[0, D)$, so the synaptic set for all neurons is compactly represented by the weight tensor $\mathbf{W} \in \mathbb{R}^{N \times N \times D}$, with

$$\mathbf{W}_{b_s, a_s, \delta_s} = w_s \quad \forall s \in [1,\, N_s], \quad \text{and zero otherwise.}$$

2.2 Recurrent HD-SNN architecture

The recurrent Heterogeneous-Delay SNN (HD-SNN) consists of a population $\mathcal{A}$ of $N = 1024$ leaky integrate-and-fire (LIF) neurons implemented in snnTorch [7] and simulated over $T = 1000$ steps of 1 ms each (total duration: 1 s). Extending the feedforward Spiking Motif Detector of Perrinet [20], our architecture includes full recurrence: each neuron $j \in \mathcal{A}$ integrates inputs from *all* other neurons across the preceding $D = 41$ time steps. This particular choice for the maximum delay is justified from biological observations and similar to the value used in the original polychronisation study [14]. The membrane potential $u_j(t)$ evolves as

$$u_j(t) = \beta \cdot u_j(t-1) \cdot (1 - s_j(t-1)) + \sum_{i=1}^{N} \Big(\sum_{d=1}^{D} \mathbf{W}_{j,i,d} \cdot s_i(t-d) \Big), \tag{1}$$

where $\beta \in (0, 1)$ is the membrane decay constant ($\beta = 0.7$ in our simulations, that is, a time constant of $\tau = 1/\ln(\frac{1}{\beta}) \approx 2.8$ steps), $\mathbf{W}_{j,i,d}$ is the synaptic weight from neuron i to neuron j at delay d, and $s_j(t) \in \{0, 1\}$ is the spike indicator $s_j(t) = \mathbb{1}[u_j(t) \geq \vartheta]$, that is, a spike is emitted when $u_j(t)$ exceeds the threshold $\vartheta = 1$. After spiking, the membrane potential that is accumulated in Equation 1 is reset to zero.

In our implementation, the spike history of all N neurons over the past D steps is concatenated into a flattened context vector of dimension $N \cdot D$ at each time step, so that $\mathbf{W}$ is reshaped as $\mathbf{W} \in \mathbb{R}^{N \times (N \cdot D)}$. This is equivalent to the temporal convolution that we used previously in a feedforward setting [20], here extended to a fully recurrent topology. Using such an implementation, the HD-SNN is thus corresponding to a standard computational graph which we could train using surrogate gradient descent (see Figure 2 - Left).

2.3 Synthetic dataset and memory cuing

To assess working-memory capacity, we generated a dataset of $M = 16$ distinct target patterns $\mathbf{S}^{(\mu)} \in \{0, 1\}^{N \times T}$. Following Perrinet [20], each pattern is drawn as independent Bernoulli trials over N neurons and T time steps with

$$\mathbf{S}^{(\mu)} \sim \sigma(\sigma^{-1}(p_A) + e_A^{(\mu)})$$

for $\mu \in [1 \ldots M]$ and where p_A is the mean firing rate that we set as 1 Hz (spike probability 1×10^{-3} per neuron per step) and the evidence term ${e_A}^{(\mu)} \in \{0, 1\}^{N \times T}$ is first drawn from a normal distribution of standard deviation $E_{SM} = 4$:

$${e_A}^{(\mu)} \sim \mathcal{N}(0, e)$$

To simulate realistic sparse activity, we retain only the top $p_{SM} = 0.5\%$ of these activation values, setting the rest to zero. Finally, a refractory period is enforced by imposing a minimum inter-spike interval of 4 ms, yielding biologically plausible sparse activity.

Given a target pattern $\mathbf{S}^*$, each contiguous window of length D, i.e. the slice $\mathbf{S}^*[\cdot, t-D : t-1]$, constitutes a unique context that may predict the spiking activity of all neurons at time t. The network is therefore trained to generate a sequential chain of overlapping Spiking Motifs [15, 20] that faithfully reproduces the target. Memory retrieval is cued by triggering all neuron activities to the target pattern during the initial D time steps ($0 \leq t < D$), corresponding to one elementary Spiking Motif (see Figure 2 - right); the network then generates the remainder of the sequence autonomously.

2.4 Analytical weight initialisation

Before training, the weight tensor $\mathbf{W}$ can be initialised analytically by treating the task as a linear regression problem. For each target pattern $\mathbf{S}^{*(\mu)} \in \{0,1\}^{N\times T}$, the context window at time t is the flattened vector $\mathbf{c}^{(\mu)}(t) \in \{0,1\}^{N\cdot D}$ containing the spike history of all neurons over steps $t-D$ to $t-1$. Stacking all $K = M\cdot(T-D)$ context–output pairs across patterns and time steps yields

$$\mathbf{C} \in \{0,1\}^{(N\cdot D)\times K}, \qquad \mathbf{I}^* \in \mathbb{R}^{N\times K},$$

and the initialisation problem is $\mathbf{W}\,\mathbf{C} \approx \mathbf{I}^*$.

The choice of regression target $\mathbf{I}^*$ requires care. A naive choice is the binary spike matrix $\mathbf{S}^*$ itself, which would ask the network to produce a current equal to the target spike at each step. However, from Equation 1, the synaptic current $I_j(t) = \sum_{i,d} \mathbf{W}_{j,i,d} \cdot s_i(t-d)$ does not directly generate a spike: it is first *filtered* by the membrane which acts as a causal first-order IIR low-pass filter with transfer function $H(z) = \frac{1}{1-\beta\cdot z^{-1}}$. To produce a membrane potential that rises precisely to $\vartheta = 1$ at t^* and is immediately reset (a temporal Dirac), the required input current is the deconvolution of this target membrane trajectory by $h(t)$. The inverse filter of $H(z)$ is $H^{-1}(z) = 1 - \beta\cdot z^{-1}$, which corresponds to the first-order difference operator. Applied to the binary target spike train $s_j^*(t)$, the optimal input current is therefore

$$I_j^*(t) \;=\; s_j^*(t) - \beta\cdot s_j^*(t-1). \tag{2}$$

This is a *biphasic* waveform: $+1$ at each target spike time t^*, $-\beta$ exactly one step later (which cancels the residual membrane charge left by the leak), and zero everywhere else. For $\beta \to 0$ the correction vanishes and $I^*(t) \to s^*(t)$, recovering the naive binary target. For $\beta = 0.8$ the correction term -0.8 is substantial and significantly reduces spurious early spikes at initialisation.

Since the system is strongly underdetermined ($K \ll ND$), the minimum-ℓ_2-norm solution is the Moore–Penrose pseudo-inverse $\mathbf{W}^* = \mathbf{I}^*\cdot\mathbf{C}^+$. With sparse Bernoulli activity at rate $p_A = 10^{-3}$, consecutive context windows are nearly

orthogonal, so the Gram matrix satisfies $\mathbf{C}\mathbf{C}^\top \approx NDp_A \cdot \mathbf{I}$, giving the closed-form initialisation

$$w_{i,j,d} = \frac{1}{N \cdot D \cdot p_A \cdot M} \sum_{t=D+1}^{T} \sum_{\mu=1}^{M} s_i^{*(\mu)}(t-d) \cdot I_j^*(t). \tag{3}$$

Each weight $w_{ij}^{(d)}$ is thus proportional to the normalised cross-correlation between the *deconvolved* spike train of target neuron j and the raw spike train of neuron i shifted by delay d, averaged over all M stored patterns. This initialisation resembles classical Hebbian learning [19] and mirrors the form of spike-timing-dependent plasticity (STDP) as used in the seminal paper on polychronisation [14], with the additional $-\beta$ correction term that accounts for the membrane leak. Note, however, that at inference time the network does not receive ground-truth contexts but the context it generated itself at the previous step, so the spike train may be corrupted by previous errors.

2.5 Training

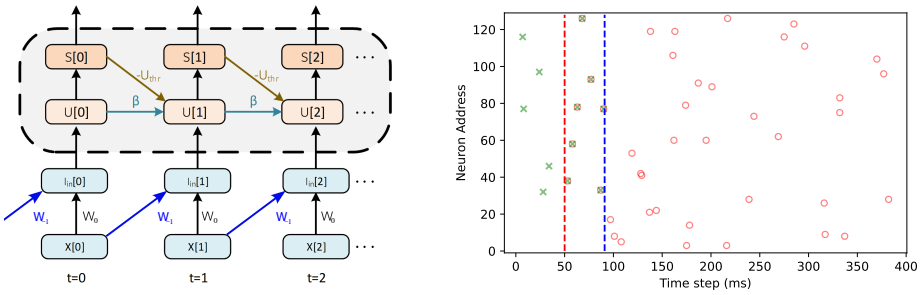

Fig. 2. Network architecture and a sample target pattern. *Left:* Unrolled view of the recurrent HD-SNN over time (one column representing one time step), illustrating delayed connections between neurons: Blue arrows indicate synaptic connections that integrate past activity across heterogeneous delays; for clarity only one delay is shown. *Right:* One representative target spike pattern ($N = 1024$ neurons, $T = 1000$ time steps, of which we show 256 neurons and 400 time steps); each red circle marks one target spike. The goal of the network is to memorise $M = 16$ such patterns, where input spikes (green crosses) are random for the first 50 steps and then we define a trigger window by activating the first $D = 41$ time steps of the target (as denoted by green crosses in the region between the dashed lines), the recurrent network predicting the following spikes.

The loss is defined as $\mathcal{L} = 1 - F_1$, where F_1 is the mean F1 score between the predicted and target spike trains, averaged over time (in the prediction window), neurons and patterns. The F1 score is the harmonic mean of precision (proportion of produced spikes that are correct) and recall (proportion of target spikes that are successfully reproduced), so minimising $\mathcal{L}$ simultaneously penalises

over-active networks (low precision) and silent networks (low recall). Learning is carried out with surrogate-gradient BPTT [17], using a fast-sigmoid surrogate with sharpness parameter $\alpha = 15$. Weights are optimised with SGD (learning rate $\eta = 10^{-3}$, momentum $\mu = 0.99$, weight decay $\lambda = 0$). Dropout regularization with probability 0.37 is applied. Training runs for 16 epochs; to mitigate overfitting and stabilise delay learning, the learning rate is annealed with a cosine schedule with warmup, as used for instance in [12]. The validation loss is monitored for detecting overfitting.

The gradient-based learning in our recurrent HD-SNN shares fundamental principles with biological STDP, despite operating at different scales. Both mechanisms exploit temporal correlations in spike timing to adjust synaptic strengths. However, while STDP implements local pairwise updates based on individual spike timings, our approach performs global optimisation that effectively learns the collective interaction of multiple delayed spike trains across the entire network. The analytical initialisation (see Equation 3) reveals this connection most clearly: it implements a Hebbian-like rule that is mathematically equivalent to averaging STDP updates across all stored patterns, suggesting that gradient descent in this context can be viewed as a principled, globally optimal implementation of temporally structured synaptic plasticity.

Our implementation is available at https://github.com/laurentperrinet/MNESIS. The recurrent HD-SNN was implemented using the snnTorch [7] library with custom extensions for heterogeneous delay handling. The implementation leverages PyTorch with the Metal Performance Shader (MPS) backend for accelerated computation on Apple Silicon (M3 Ultra chip featuring 32 CPU cores and 80 GPU cores, with 512 GB of unified memory architecture) and further tested with the CUDA backend on the Jean Zay supercluster at GENCI. The MPS and CUDA backends enabled efficient training and simulation of the large recurrent networks required for working memory tasks, leveraging the parallel processing capabilities of the GPU cores while maintaining low-latency access to the substantial memory resources necessary for storing all tensors' parameters. Training typically converged within 128 gradient steps, requiring approximately 5 minutes of computation time per training.

3 Results

3.1 Training dynamics and results

Performing training of the synthetic data on the recurrent HD-SNN model with surrogate-gradient BPTT [17] consistently maximised the mean F1 score across all $M = 16$ target patterns to a perfect score of 1.0, with $N = 1024$ neurons, $D = 41$ delays, $T = 1000$ time steps, $p_A = 10^{-3}$, and $\beta = 0.7$ (time constant $\tau \approx 2.8$ ms). When using random intialization instead of the procedure described in Section **??**, we observed two characteristic failure modes during early training: episodes of complete silence, in which the network emits no spikes, and over-activation, in which nearly all neurons fire at every time step. Both are well-known pathological attractors of recurrent SNNs [14]. The F1 score is the harmonic

mean of precision and recall, and maximising it therefore penalises both failure modes symmetrically: low precision corresponds to an over-active network and low recall to a silent one. The cosine learning-rate schedule [12] helped guide the network away from these extremes towards a balanced, sparse-firing regime.

Remarkably, the analytical weight initialisation alone (Section 2, Eq. 3) already achieves $F_1 = 1.0$ before any gradient step, demonstrating that the closed-form solution is sufficient for perfect recall on this synthetic benchmark. We also validated that the Hebbian approximation of the initialisation of weights yielded a perfect F1-score. This greatly improved the computing time (by a factor of ≈ 5) as it involved only a simple correlation instead of the computation of the pseudo-inverse. We kept this method in the remainder of this work. It should be noted that we noticed that errors occur when introducing noise (see below) and that learning helped mitigate such issues.

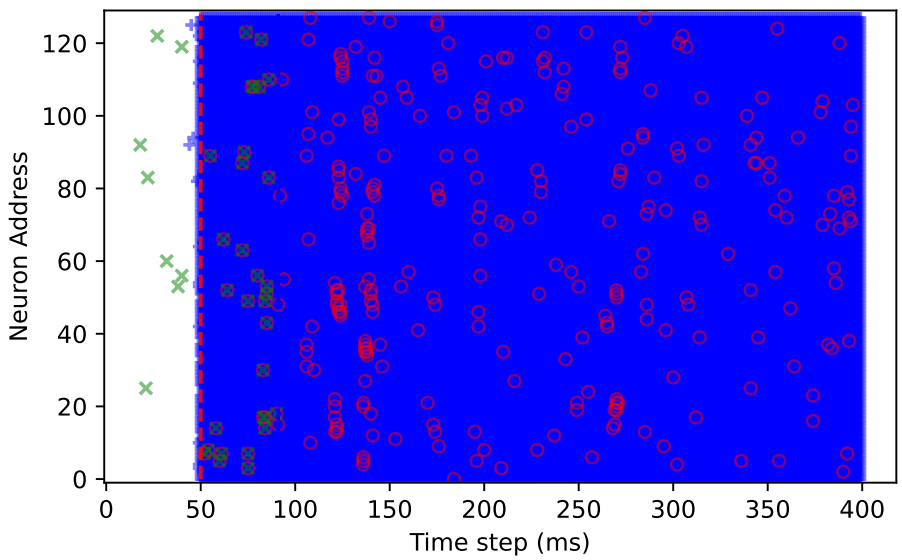

Fig. 3. Pattern retrieval from the recurrent HD-SNN. The raster plot shows network activity (blue crosses, $N = 1024$ neurons) versus one of the $M = 16$ target patterns (red circles) after training. Correct recall emerges first in the time steps immediately following the trigger window ($0 \leq t < D = 41$, delimited by the dashed red line) and then progressively extends to the full sequence duration ($T = 1000$ steps).

3.2 Effect of hyperparameters

We performed a systematic sweep over the key hyperparameters and report their effect on the mean F1 score at convergence.

Surrogate sharpness α. The sharpness of the surrogate gradient had the largest single effect on performance. Low values of α produced gradients that were too smooth to discriminate precise spike times, while excessively high values caused gradient vanishing. The value $\alpha = 10$ provided the best trade-off, consistent with findings in Neftci et al. [17].

Target firing rate. The best recall was obtained at low target firing rates (1 Hz per neuron), confirming that sparse activity facilitates stable recurrent dynamics [14, 28]. Higher firing rates led to increased inter-pattern interference and degraded precision, echoing the capacity bounds reported for polychronous neuronal groups [24].

3.3 Retrieval of multiple spike patterns

Training was not uniform in time: the network first reproduced spikes immediately following the trigger window ($0 \leq t < D$), then progressively extended correct recall towards later time steps, as shown in Figure 3. This behaviour is a direct consequence of the sequential structure of the task: because each context window $[t - D,\, t - 1]$ uniquely predicts the activity at time t, errors at early time steps propagate into all subsequent predictions, making temporal proximity to the trigger window the primary driver of learning progress.

To evaluate the network's ability to retrieve and distinguish multiple stored patterns in sequence, we concatenated all $M = 16$ patterns into a single long raster separated by inter-trial intervals of spontaneous background activity (Figure 4). Each pattern was triggered by its individual trigger window, and retrieval was evaluated by a sliding-window F1 score over time. The F1 score rose sharply at the onset of each trigger window, remained high throughout each pattern's duration, and dropped back to the background level during the inter-trial intervals, confirming that the network correctly associates each trigger window with its corresponding stored pattern without cross-interference between the $M = 16$ stored memories.

3.4 Working memory and robustness to perturbations

A hallmark of biological working memory is graceful degradation: recall should remain possible under incomplete or corrupted retrieval cues, mirroring the tolerance of hippocampal pattern completion to partial input [26]. We therefore evaluated three distinct perturbations of the trigger window, each probing a different dimension of robustness (Figure 5).

Robustness to spike noise (Figure 5, left). We corrupted the trigger window by independently flipping each spike with probability $p_{\text{flip}} \in [0, 1]$ (bit-flip noise). At $p_{\text{flip}} = 0.25$, the network still achieved precision $= 1.000$, recall $= 0.937$, and $F_1 = 0.967$, demonstrating high tolerance to noise in the retrieval cue. The F1 score decreases monotonically as p_{flip} increases, reaching chance level only near $p_{\text{flip}} = 0.5$ where the trigger window is entirely randomised. This behaviour is

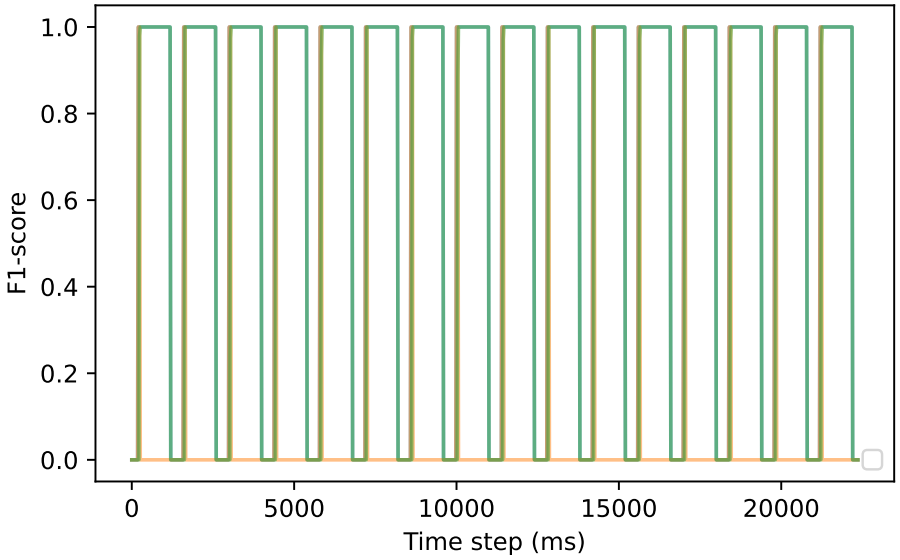

Fig. 4. Sequential retrieval of $M = 16$ stored patterns. The sliding-window F1 score (blue) is shown as a function of time, together with the onset indicator of each trigger window (green dashes) and the corresponding pattern mask (red dashes). The F1 score rises sharply at each trigger-window onset and remains high for the full pattern duration ($T = 1000$ steps), then returns to baseline during the inter-trial intervals, demonstrating selective retrieval of all stored memories in sequence without cross-interference.

consistent with the network operating as an attractor: small perturbations of the initial state are corrected by the recurrent dynamics, which converge back to the stored pattern.

Robustness to trigger-window duration (Figure 5, middle). We varied the duration of the trigger window from 0 to $D - 1 = 40$ steps, scanning from no cue at all to a trigger that is one step short of the full motif. At 75% of the full trigger duration (≈ 30 steps out of $D = 41$), the network achieved precision $= 1.000$, recall $= 0.757$, and $F_1 = 0.862$. The F1 score increases monotonically with trigger duration, rising steeply once the trigger exceeds roughly $D/2$, consistent with the requirement that the context window must contain enough active synapses to uniquely identify the stored pattern above the interference floor.

Robustness to partial neuron coverage (Figure 5, right). We varied the number of neurons whose activity was silenced during the trigger window, from 0 (full trigger) to $N = 1024$ (no trigger). With $N/8 = 128$ neurons silenced — meaning 87.5% of the population remained active — the network still achieved perfect recall ($F_1 = 1.000$). The F1 score degrades smoothly as more neurons are silenced

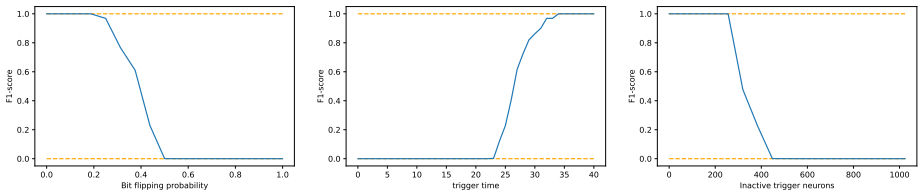

Fig. 5. Robustness of working-memory retrieval to three types of trigger-window perturbation (F_1 score averaged over $M = 16$ patterns). *Left:* Bit-flip noise applied to all spikes in the trigger window ($p_{\text{flip}} \in [0, 1]$); the network maintains high F_1 up to $p_{\text{flip}} \approx 0.4$. *Middle:* Truncated trigger window (trigger duration from 0 to $D - 1 = 40$ steps); retrieval remains reliable for durations above roughly $D/2$. *Right:* Partial neuron coverage (number of silenced neurons from 0 to $N = 1024$); perfect recall is maintained until the majority of the trigger population is inactive. Orange dashed lines indicate $F_1 \in \{0, 1\}$.

and drops sharply only when the majority of the trigger population is inactive, again consistent with an attractor interpretation in which a sufficient fraction of the pattern suffices to pull the network dynamics towards the stored memory.

3.5 Effect of delay depth, pattern duration, and background firing rate

To characterise how the architectural parameters govern working-memory performance, we conducted a systematic one-at-a-time scan over the maximum delay D, the pattern duration T, and the background firing rate p_A, holding all other hyperparameters at their default values (Section 2). Each condition was evaluated with $N_{\text{cv}} = 10$ independent random seeds and results are reported as mean loss $\mathcal{L} = 1 - F_1$ (Figure 6).

Delay depth D (Figure 6, left). The loss decreases monotonically as D grows from 3 to 127 ms, dropping from $\mathcal{L} \approx 0.85$ at $D = 3$ to near zero at $D = 127$. This confirms the capacity argument: the parameter budget $N^2 \cdot D$ and the context-window size $N \cdot D$ both scale linearly with D. A wider delay window provides more temporally diverse synaptic pathways, increasing the number of simultaneously representable Spiking Motifs and reducing interference between overlapping context windows. At $D = 3$ the context vector has only $N \cdot D = 3072$ dimensions with expected activity $N \cdot D \cdot p_A \approx 3.07$ — a regime where contexts are nearly all-zero and indistinguishable. At $D = 127$ the context has $N \cdot D = 129{,}024$ dimensions with expected activity ≈ 129, and near-orthogonality is strongly satisfied.

Pattern duration T (Figure 6, middle). The loss grows monotonically with T, from $\mathcal{L} \approx 0.004$ at $T = 64$ to $\mathcal{L} \approx 0.08$ at $T = 2048$. This is the empirical signature of the two compounding difficulties identified in the capacity analysis: longer

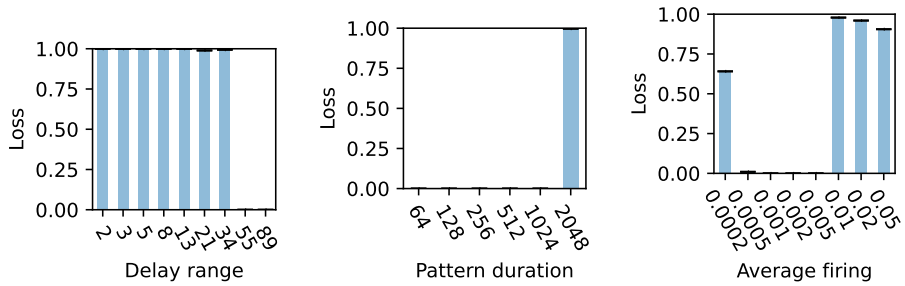

Fig. 6. Effect of three key parameters on working-memory performance (loss $\mathcal{L} = 1 - F_1$, mean $\pm$ s.e. over $N_{\mathrm{cv}} = 10$ cross-validation folds). *Left:* Increasing the maximum delay D consistently reduces the loss, confirming that deeper delay windows provide a larger, more orthogonal context and that longer maximum delays directly increase the capacity of the recurrent HD-SNN. *Middle:* Loss grows slowly and monotonically with pattern duration T, consistent with the compounding-error and gradient-vanishing arguments: longer sequences require the network to maintain coherent predictions further from the trigger window. *Right:* Performance is optimal at low background firing rates ($p_A \lesssim 10^{-3}$) and degrades sharply at higher rates.

sequences impose a larger open-loop rollout, and the error propagation recurrence accumulates over $T - D$ steps. Even a sub-critical propagation coefficient $\lambda < 1/D$ eventually allows errors to reach a non-negligible fraction of N spikes.

Background firing rate p_A (Figure 6, right). The loss exhibits a clear optimum around $p_A = 10^{-4}$ to 10^{-3} and degrades rapidly for $p_A \geq 2 \times 10^{-3}$. Increasing p_A linearly reduces the SNR of the cross-correlation initialisation and increases the Gram-matrix off-diagonal terms, degrading context orthogonality.

Taken together, the three scans quantitatively validate the theoretical framework: delay depth is the primary capacity lever (loss $\propto 1/(N \cdot D \cdot p_A)$), pattern duration is the primary difficulty lever, and firing rate sets the orthogonality regime that determines whether initialisation and learning converge. These results suggest that the most efficient path to storing longer patterns is to increase D rather than N, since D enters both the parameter capacity $N^2 \cdot D$ and the context orthogonality $N \cdot D \cdot p_A$ while adding no new neurons or lateral connections. Increasing D is, however, ultimately limited by biological constraints on axonal conduction delays [10].

4 Discussion

4.1 Memory capacity and the role of heterogeneous delays

We have demonstrated that a recurrent SNN with heterogeneous synaptic delays can store and reproduce arbitrary target spike patterns, with the analytical weight initialisation alone achieving perfect recall ($F_1 = 1.0$) on a benchmark of $M = 16$

patterns of $T = 1000$ steps. The result is noteworthy from a capacity standpoint: the entire network is parameterised by a single weight tensor $\mathbf{W} \in \mathbb{R}^{N \times N \times D}$ of dimension $N^2 \cdot D = 1024^2 \cdot 41 \approx 43 \times 10^6$ parameters, yet it faithfully stores M patterns each of $N \cdot T$ binary values. This compression is made possible precisely by the temporal structure introduced by the D delay channels: rather than requiring one parameter per stored bit, the network exploits the sequential dependencies between consecutive context windows of length D to represent the entire pattern as a chain of overlapping Spiking Motifs [20**?**]. The memory limit is determined by parameter count rather than interference — the marginal parameter cost per additional stored pattern is zero, and the capacity scales as $N^2 \cdot D$, linear in the delay depth. This provides concrete evidence that heterogeneous delays are a computational asset rather than a biological nuisance, a point argued theoretically in Izhikevich [14] and demonstrated empirically in the context of learnable delays in feedforward [11, 12] and recurrent architectures [16, 21], and which we extend here to a full working-memory setting.

The robustness experiments further strengthen this interpretation. The network tolerates up to 25% bit-flip noise in the seed window, requires only 75% of the full seed duration for reliable recall, and maintains perfect retrieval with up to 87.5% of seed neurons silenced. This progressive degradation profile is consistent with the network operating as an attractor: the stored patterns correspond to stable fixed points of the recurrent dynamics, and the seed window provides a partial cue that the network completes autonomously — the same mechanism proposed for hippocampal pattern completion [26].

4.2 Limitations and directions for improvement

The present model is deliberately minimal, and several design choices impose limits on capacity and robustness that future work should address.

Capacity and scaling. A systematic evaluation of memory capacity — how many patterns of what duration can be reliably stored and retrieved — remains to be carried out beyond the present $M = 16$ benchmark. The theoretical analysis predicts that capacity scales as $N^2 \cdot D/(T - D)$, giving approximately 5,700 patterns for the current architecture, but whether gradient-based training reliably reaches this bound at scale is an open empirical question. Interference between stored patterns at high memory load may degrade precision in ways that the F1 training objective does not fully penalise, and the sensitivity of recall to perturbations of the seed window is expected to decrease as M grows.

Richer neuron models and dynamics. We used the simplest LIF model with a fixed membrane decay β. Introducing adaptive threshold dynamics or more complex single-neuron models, as studied in Szatmáry and Izhikevich [24], should increase the diversity of intrinsic timescales available to the network and improve both capacity and retrieval fidelity. In particular, the deconvolution analysis (Section 2) shows that the optimal weight profile depends critically on β: larger values of β require exponentially amplified weights at long delays, suggesting that

adaptive β per neuron could allow the network to exploit the full delay range more efficiently.

Latent reservoir neurons. The current architecture connects every neuron directly to every other neuron across all delays. Partitioning the population into input-output neurons and a larger pool of latent "reservoir" neurons, as in delay-based reservoir computing [18], would enrich the internal spiking dynamics without increasing the number of trained parameters proportionally. Such a hybrid architecture could combine the representational power of random recurrent networks with the precision of gradient-trained delay weights [17], potentially closing the gap between biologically inspired models and sparse selective-update architectures [28].

Online and unsupervised learning. The present model requires a fixed set of target patterns known in advance. A natural extension is an online learning rule that updates the weights continuously as new patterns are presented, without catastrophic forgetting of previously stored sequences. The closed-form Hebbian initialisation (Eq. 3) suggests a natural online update: each new spike $s_j(t)$ incrementally adjusts $w_{ij}^{(d)}$ in proportion to $s_i(t-d)$, recovering a local STDP-like rule [14]. Whether such online updates are stable under recurrent feedback and can support continual working memory remains to be established.

4.3 Perspectives: unsupervised learning and neural data

The most immediate application of this framework is to neurophysiological data. The supervised memorisation task studied here requires a known target pattern, but in practice one would like to discover the latent Spiking Motifs present in recorded multi-unit activity — the problem originally addressed by Ikegaya et al. [13] and Villette et al. [26]. A natural extension is to replace the supervised F1 loss with a self-supervised or contrastive objective that trains the network to predict upcoming spikes from the recent spike history, without access to ground-truth labels. This is analogous to the contrastive predictive coding paradigm applied to continuous neural time series, here instantiated in a fully spike-based setting [10]. Such a model would provide a principled, energy-efficient method for identifying the temporal structure of neural population codes, with direct relevance to the debate on rate coding versus temporal coding [4, 23] and to understanding the neural mechanisms of working memory more broadly. Deployed on neuromorphic hardware [7, 8], it could operate in real time on streaming electrophysiological data, opening a new avenue for closed-loop brain-machine interface applications.

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References

1. Abeles, M.: Role of the cortical neuron: Integrator or coincidence detector? Israel Journal of Medical Sciences **18**(1), 83–92 (Jan 1982), ISSN 0021-2180
2. Adrian, E.D., Zotterman, Y.: The impulses produced by sensory nerve endings. The Journal of Physiology **61**(4), 465–483 (Aug 1926), ISSN 0022-3751, URL https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1514868/
3. Bienenstock, E.: A model of neocortex. Network: Computation in Neural Systems **6**(2), 179–224 (Jan 1995), ISSN 0954-898X
4. Bohte, S.M.: The evidence for neural information processing with precise spike-times: A survey. Natural Computing **3**(2), 195–206 (2004), https://doi.org/10.1023/B:NACO.0000027755.02868.60
5. Bruno, R.M., Sakmann, B.: Cortex Is Driven by Weak but Synchronously Active Thalamocortical Synapses. Science **312**(5780), 1622–1627 (2006), https://doi.org/10.1126/science.1124593, URL https://doi.org/10.1126/science.1124593
6. Eguchi, A., Isbister, J.B., Ahmad, N., Stringer, S.: The emergence of polychronization and feature binding in a spiking neural network model of the primate ventral visual system. Psychological Review **125**(4), 545–571 (2018), ISSN 1939-1471, https://doi.org/10.1037/rev0000103, URL https://psycnet.apa.org/fulltext/2018-25960-001.pdf
7. Eshraghian, J.K., Ward, M., Neftci, E., Wang, X., Lenz, G., Dwivedi, G., Bennamoun, M., Jeong, D.S., Lu, W.D.: Training spiking neural networks using lessons from deep learning. Proceedings of the IEEE **111**(9), 1016–1054 (2023), https://doi.org/10.1109/JPROC.2023.3308088
8. Göltz, J., Kriener, L., Baumbach, A., Billaudelle, S., Breitwieser, O., Cramer, B., Dold, D., Kungl, A.F., Senn, W., Schemmel, J., Meier, K., Petrovici, M.A.: Fast and energy-efficient neuromorphic deep learning with first-spike times. Nature Machine Intelligence **3**(9), 823–835 (Sep 2021), ISSN 2522-5839, https://doi.org/10/gm5gwd, URL https://www.nature.com/articles/s42256-021-00388-x
9. Grimaldi, A., Boutin, V., Ieng, S.H., Benosman, R., Perrinet, L.U.: A robust event-driven approach to always-on object recognition. Neural Networks **178**, 106415 (2024), https://doi.org/10.1016/j.neunet.2024.106415, URL https://www.sciencedirect.com/science/article/pii/S0893608024003393
10. Grimaldi, A., Gruel, A., Besnainou, C., Jérémie, J.N., Martinet, J., Perrinet, L.U.: Precise Spiking Motifs in Neurobiological and Neuromorphic Data. Brain Sciences **13**(1), 68 (Jan 2023), ISSN 2076-3425, https://doi.org/10.3390/brainsci13010068, URL https://www.mdpi.com/2076-3425/13/1/68

11. Grimaldi, A., Perrinet, L.U.: Learning heterogeneous delays in a layer of spiking neurons for fast motion detection. Biological Cybernetics **117**, 373–387 (Sep 2023), https://doi.org/10.1007/s00422-023-00975-8, URL https://link.springer.com/article/10.1007/s00422-023-00975-8
12. Hammouamri, I., Khalfaoui-Hassani, I., Masquelier, T.: Learning Delays in Spiking Neural Networks using Dilated Convolutions with Learnable Spacings (Jun 2023), https://doi.org/10.48550/arXiv.2306.17670, URL http://arxiv.org/abs/2306.17670
13. Ikegaya, Y., Aaron, G., Cossart, R., Aronov, D., Lampl, I., Ferster, D., Yuste, R.: Synfire Chains and Cortical Songs: Temporal Modules of Cortical Activity. Science **304**(5670), 559–564 (Apr 2004), https://doi.org/10.1126/science.1093173, URL https://www.science.org/doi/10.1126/science.1093173
14. Izhikevich, E.M.: Polychronization: Computation with Spikes. Neural Computation **18**(2), 245–282 (Feb 2006), ISSN 0899-7667, https://doi.org/10.1162/089976606775093882, URL https://doi.org/10.1162/089976606775093882
15. Kronland-Martinet, T., Viollet, S., Perrinet, L.U.: Detection of spiking motifs of arbitrary length in neural activity using bounded synaptic delays (2025), https://doi.org/10.48550/ARXIV.2511.15296, URL https://arxiv.org/abs/2511.15296
16. Mészáros, B., Knight, J.C., Nowotny, T.: Efficient Event-based Delay Learning in Spiking Neural Networks (Jun 2025), https://doi.org/10.48550/arXiv.2501.07331, URL http://arxiv.org/abs/2501.07331
17. Neftci, E.O., Mostafa, H., Zenke, F.: Surrogate Gradient Learning in Spiking Neural Networks: Bringing the Power of Gradient-Based Optimization to Spiking Neural Networks. IEEE Signal Processing Magazine **36**(6), 51–63 (Nov 2019), ISSN 1053-5888, https://doi.org/10.1109/MSP.2019.2931595, URL https://ieeexplore.ieee.org/document/8891809/
18. Paugam-Moisy, H., Martinez, R., Bengio, S.: Delay learning and polychronization for reservoir computing. Neurocomputing **71**(7–9), 1143–1158 (Mar 2008), ISSN 0925-2312, https://doi.org/10.1016/j.neucom.2007.12.027, URL http://www.sciencedirect.com/science/article/pii/S0925231208000507
19. Perrinet, L., Samuelides, M.: Coherence detection in a spiking neuron via Hebbian learning. Neurocomputing **44–46**, 133–139 (2002), https://doi.org/10.1016/s0925-2312(02)00374-0
20. Perrinet, L.U.: Accurate Detection of Spiking Motifs in Multi-unit Raster Plots. In: Iliadis, L., Papaleonidas, A., Angelov, P., Jayne, C. (eds.) Artificial Neural Networks and Machine Learning — ICANN 2023, pp. 369–380, Springer Nature Switzerland, Cham (2023), ISBN 978-3-031-44206-3, https://doi.org/10.1007/978-3-031-44207-0_31
21. Queant, A., Rançon, U., Cottereau, B.R., Masquelier, T.: DelRec: Learning delays in recurrent spiking neural networks (Sep 2025), https://doi.org/10.48550/arXiv.2509.24852, URL http://arxiv.org/abs/2509.24852

22. Rockland, K.S.: Non-uniformity of extrinsic connections and columnar organization. Journal of Neurocytology **31**(3-5), 247–253 (2002), ISSN 0300-4864, https://doi.org/10.1023/a:1024169925377
23. Softky, W.R., Koch, C.: The highly irregular firing of cortical cells is inconsistent with temporal integration of random EPSPs. Journal of Neuroscience **13**(1), 334–350 (Jan 1993), https://doi.org/10.1523/jneurosci.13-01-00334.1993, URL https://doi.org/10.1523/JNEUROSCI.13-01-00334.1993
24. Szatmáry, B., Izhikevich, E.M.: Spike-Timing Theory of Working Memory. PLoS Computational Biology **6**(8), e1000879 (Aug 2010), ISSN 1553-7358, https://doi.org/10.1371/journal.pcbi.1000879, URL https://dx.plos.org/10.1371/journal.pcbi.1000879
25. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I.: Attention is All you Need. NIPS pp. 5998–6008 (2017), https://doi.org/10.48550/arXiv.1706.03762, URL https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html
26. Villette, V., Malvache, A., Tressard, T., Dupuy, N., Cossart, R.: Internally Recurring Hippocampal Sequences as a Population Template of Spatiotemporal Information. Neuron **88**(2), 357–366 (Oct 2015), ISSN 0896-6273, https://doi.org/10.1016/j.neuron.2015.09.052, URL https://www.sciencedirect.com/science/article/pii/S0896627315008417
27. Xie, W., Wittig, J.H., Chapeton, J.I., El-Kalliny, M., Jackson, S.N., Inati, S.K., Zaghloul, K.A.: Neuronal sequences in population bursts encode information in human cortex. Nature **635**(8040), 935–942 (Nov 2024), ISSN 0028-0836, 1476-4687, https://doi.org/10.1038/s41586-024-08075-8, URL https://www.nature.com/articles/s41586-024-08075-8
28. Yin, B., Wang, S., Tan, H., Bohte, S., Corradi, F., Li, G.: Efficient Sparse Selective-Update RNNs for Long-Range Sequence Modeling (Feb 2026), https://doi.org/10.48550/arXiv.2603.02226, URL http://arxiv.org/abs/2603.02226